The Entropy Cube of the Heart

Structured Knowledge Accumulation on the Raw ECG Stream — Research Proposal

Bouarfa Mahi, Quantiota

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Abstract

The electrocardiogram is usually reduced to positions, intervals and the R–R series before any analysis. This proposal follows a different path: the raw ECG is read as an ordered stream of waves, P $\rightarrow$ Q $\rightarrow$ R $\rightarrow$ S $\rightarrow$ T, and each wave is one step of the Structured Knowledge Accumulation (SKA) real-time learner [1, 2]. The learner is not trained on labels. It reduces its own entropy only where the stream carries structure, and the five transitions of the cardiac cycle form their own bands in the learner's space.

Across public PhysioNet databases, the learner registers the five transitions of healthy hearts, the time it needs to saturate its entropy (τ_H, N_H), and, on a long Holter recording ending in ventricular fibrillation, loops where the structure of the cycle is lost. We propose the **entropy cube** $(z, \dot{z}, H)$ as the standard object of a heart, recorded under fixed conditions, and a monitoring set-up that learns one loop per hour or per day, so that the structure of a heart can be followed over time.

All results cited here are public, with their cubes, in the repositories listed in Section 9.

1. Background

SKA. Structured Knowledge Accumulation is a forward-only learning framework in which learning is the reduction of an entropy, defined on the learner's knowledge z and its decision probabilities [1]. Learning follows a principle of entropic least action, and the time to complete it defines a characteristic time [2]. The real-time learner used here reads one value per step, in order, and its entropy H is recorded at every step.

Why the raw waveform. Classical heart rate variability reduces the ECG to the R–R intervals. Here the learner reads the full cycle: every wave is a step, and the rhythm appears in the time between steps rather than being extracted beforehand.

Following Wheeler. *"No elementary phenomenon is a phenomenon until it is a registered (observed) phenomenon"* [10]. The transition bands are not values present in the ECG: they are registered by the interaction between the ordered stream of waves and the learner. They are not in a histogram of the input, which discards the order of the waves, and they form only as the learner's entropy falls.

2. Method

Wave detection. For each cycle, the peaks of P, Q, R, S, T are located on the raw signal. R is found with a Pan–Tompkins-type detector [6] (50 Hz notch, 5–20 Hz band, squared slope, 150 ms energy window); P, Q, S and T are found in windows around R. The detection was checked against the cardiologists' annotations of the LUDB database [9], 200 records, lead II (Table 1, Figure 1).

Table 1: Detection against the LUDB annotations. False detections are counted in records where the wave is annotated. In 23 records with atrial fibrillation or flutter no P is annotated, and the detector still reports a P in each cycle. [a] The detector returns a cycle only between two R peaks, so the first and last QRS of each 10-s record are not returned. Of the 91 unmatched QRS, 54 lie before the first or after the last returned cycle, 21 are in two records (83, 105) where only two R peaks are found, and 16 are elsewhere.

wave	annotated	matched	median error	within 20 ms	inside the annotated wave	false detections
P	1,401	96.5 %	2 ms	97.2 %	99.0 %	3 / 1,355
R	1,817	95.0 %[a]	2 ms	99.0 %	—	5 / 1,729
T	1,642	90.1 %	4 ms	90.7 %	94.2 %	86 / 1,565
Q	1,726 QRS	—	—	—	82.1 %	—
S	1,726 QRS	—	—	—	81.1 %	—

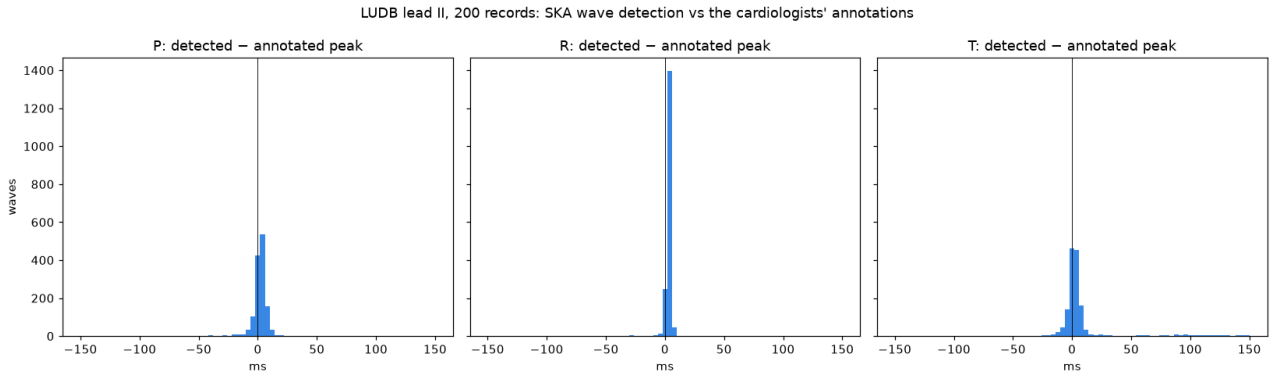


Figure 1: Detected minus annotated peak, LUDB lead II, 200 records.

Complex events. One cycle is read as one turn of a circle whose circumference is the cycle period, $T = 2\pi r$. Each wave j is summed over its arc into a phasor

$$c_j = \frac{1}{r} \sum_{t \in \mathrm{arc}_j} x(t)\, e^{\,it/r},$$

and the learner's input is the change of the phasor from one wave to the next, $\dot{c}_n = c_n - c_{n-1}$, taken as $\mathrm{Re}\,\dot{c}_n$. A wave that does not reach its expected sign is marked absent.

Learning. One step per wave, five steps per cycle. A run is 700 cycles (3,500 waves). The time between steps, Δt, is the real time between successive wave peaks. At each step the learner gives its knowledge $z = \|Z\|$, its rate $\dot{z} = dz/dt$ and its entropy H.

3. The entropy cube as the standard object of a heart

Each transition P→Q, Q→R, R→S, S→T, T→P forms its own band in the learner's space (Figure 2). Three vertical axes have been used: the transition probability P, the time Δt and the entropy H. We propose the entropy cube $(z, \dot{z}, H)$ as the standard object: it measures how the heart is structured, and the bands of the five transitions separate there.

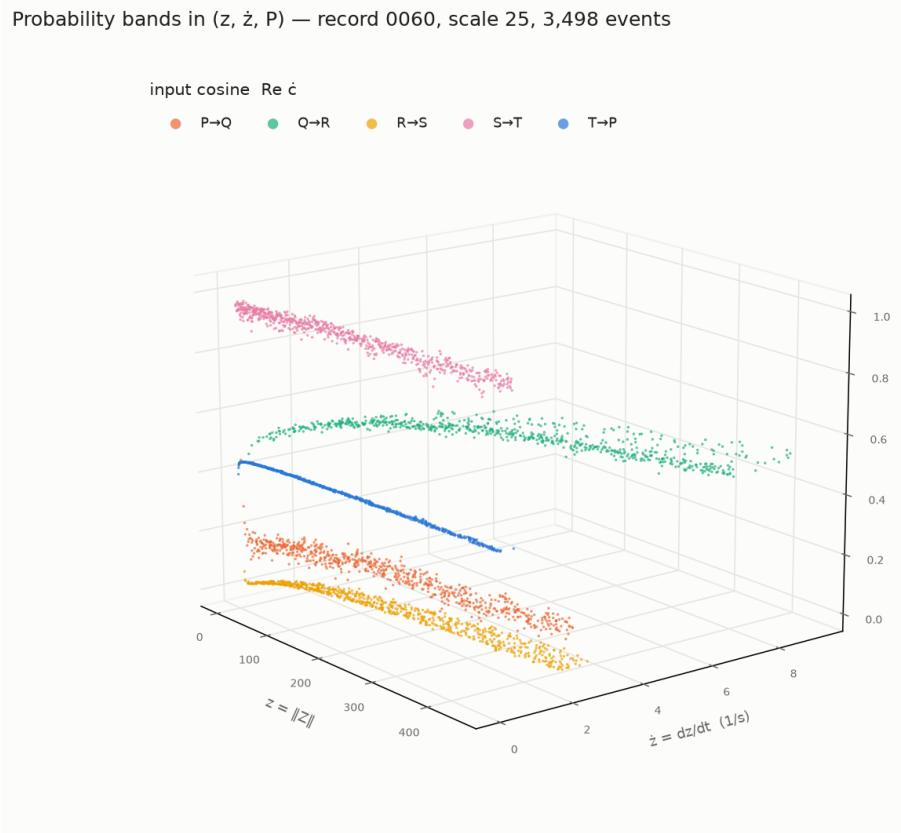


Figure 2: Record 0060 (Autonomic Aging, woman 18–19, 1000 Hz, input Re $\dot{c}$, scale 25): the five transitions as five bands in $(z, \dot{z}, P)$.

For cubes to be compared between hearts and over time, they must be recorded under the same conditions:

Table 2: The standard conditions of an entropy cube.

condition	standard
sampling frequency	1000 Hz
lead	II
input	Re $\dot{c}$
scale	25

condition	standard
run length	long enough for the entropy to saturate

The learner learns only when it finds structure. A richer structure needs more cycles before the entropy saturates, so the run length is set by saturation, not by a fixed duration.

Entropic characteristic time. τ_H is the time, $\sum_k \Delta t_k$, at which the entropy of the slowest of the five transitions is 90 % saturated. $N_H = \tau_H / T_c$, with T_c the mean cycle time of the run, removes the heart rate and counts cycles.

4. What is registered so far

Healthy women, 18 to 92 (Autonomic Aging [3], 1000 Hz). 38 women, one 700-cycle run each: 30 women, two per age band, at a scale chosen per woman before the standard above was set, and 8 more women aged 18–19 at scale 25. The five transitions form five bands in each of the 30. Over the 38, τ_H ranges from 6.7 to 13.4 min, and N_H from 515 to 695 cycles.

Healthy subjects at 250 Hz (Fantasia [4], Sport DB 2.0 [5]). 39 young and elderly subjects at supine rest, and 23 athletes at rest, learned with the same method. These recordings are at 250 Hz and are therefore outside the standard.

Myocardial infarction (PTB [7], 1000 Hz). Healthy controls and patients with myocardial infarction, from the day of the infarction to several months later, with the missing waves of each cycle tagged in the cubes. The PTB records last about two minutes: they are not sufficient for entropy saturation, and no τ_H is defined for them.

Arrhythmia: what the bands correspond to (MIT-BIH [11], 360 Hz). In the second order, the learner reads its own first-order result, one step per cycle: each step is the passage from one cycle to the next, and it carries the cardiologists' labels of both cycles. Record 119 (woman, 51) alternates normal cycles (N) and premature ventricular contractions (V), in bigeminy (N V N V) and trigeminy (N N V). Its steps form three bands that follow the labels exactly (Figure 3): N→V, the coupling interval (0.49–0.70 s, 146 steps); N→N, the normal rhythm (0.84–0.99 s, 402); V→N, the compensatory pause (1.24–1.43 s, 146). This vertical axis, Δt_2, is the time between two cycles. In the learner's own probability P_2, the coupling interval N→V forms its own band, at $P_2 \approx 0.46$, apart from the normal steps.



Figure 3: MIT-BIH record 119, second order: Δt_2 (left) and P_2 (right) of every step, coloured by the MIT-BIH labels.

Hours before and after ventricular fibrillation (SDDB [8], 250 Hz). Record 52 (woman, 82, sinus rhythm, VF onset 02:32:40), 7 h 31 min, learned as 60 successive loops of 700 cycles, each by a fresh learner at scale 25.

Table 3: N_H of the successive loops of SDDB record 52.

loops	time in the recording	N_H (cycles)
L000–L002	00:00–00:22	673–689
L003–L017	00:22–02:18	413–603
L018	02:18–02:25	624
L019 (VF onset)	02:25–02:33	416
L020–L023	02:33–02:59	396–513
L024–L059	02:59–07:30	391–694

In the first loops the five transitions keep separate entropy curves and separate levels of P (Figure 4). From loop L003, for about two hours before the fibrillation, the steps scatter between the curves and the levels mix (Figure 5): the learner saturates in fewer cycles because it finds less structure. N_H recovers in L018 (624) before falling again in L019, the loop of the VF onset. After the event, N_H varies from loop to loop over the whole range.

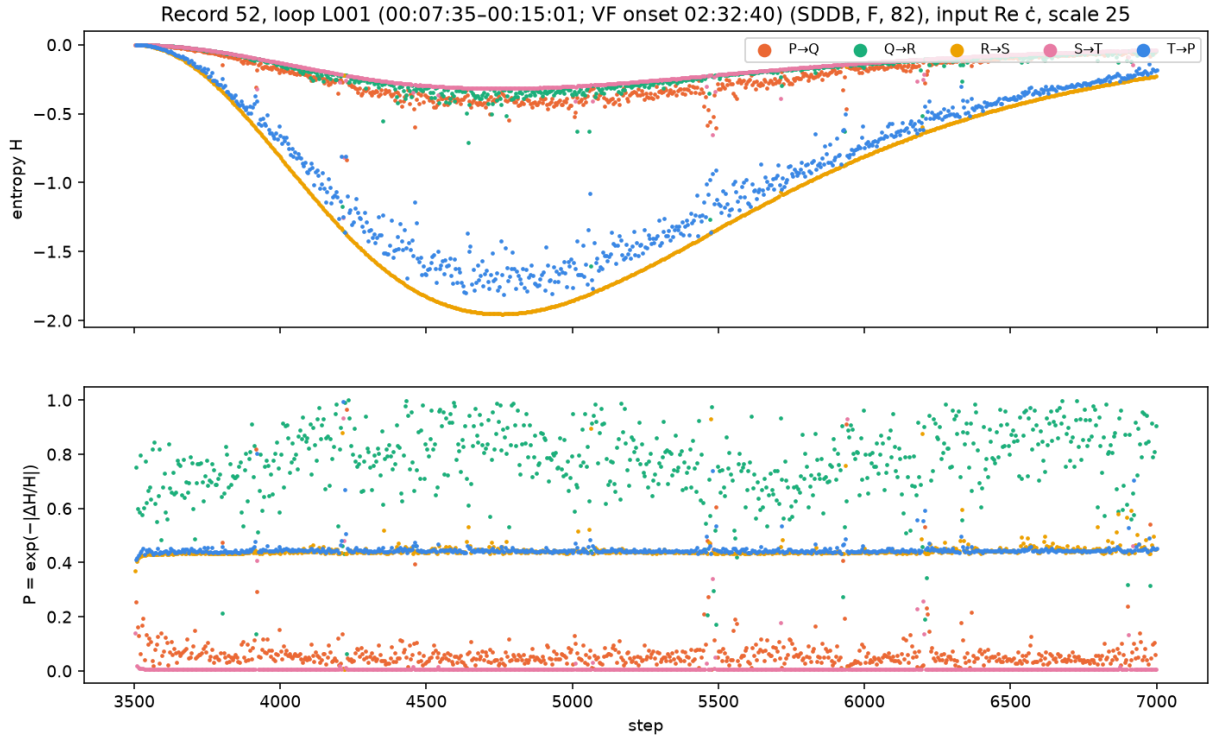


Figure 4: SDDB record 52, loop L001 (00:07–00:15), $N_H = 688$: entropy H and P per step.

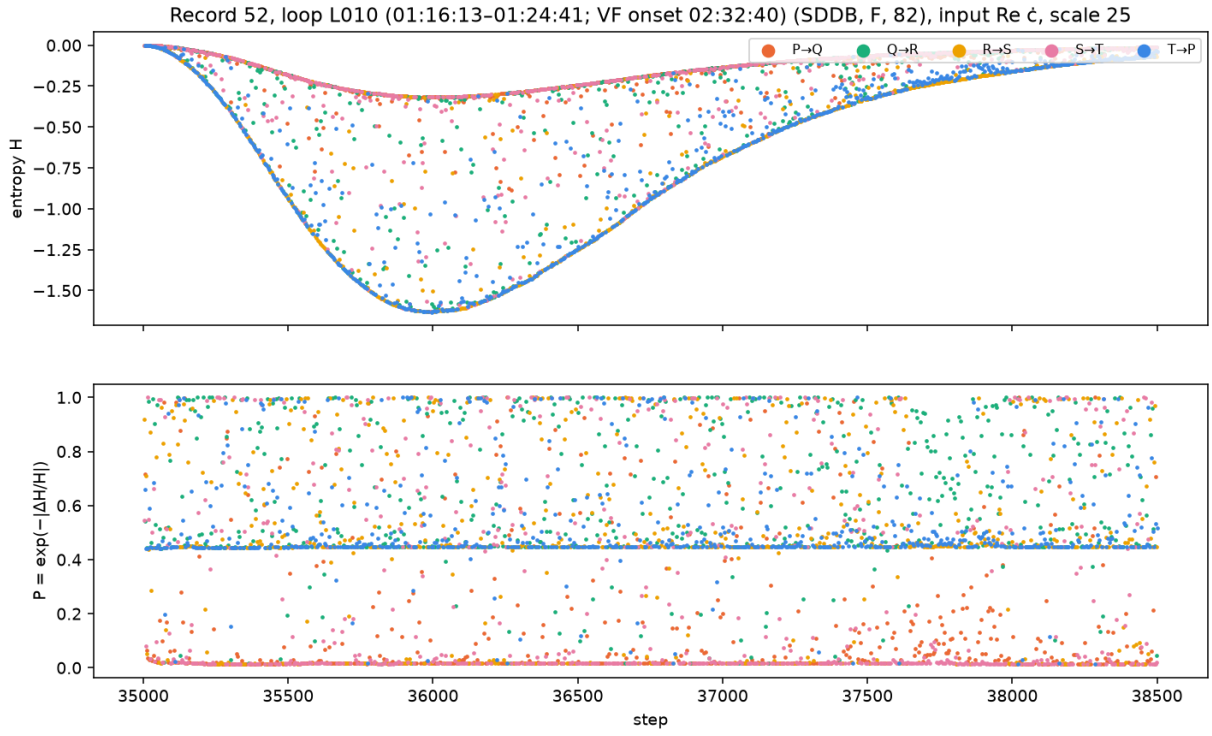


Figure 5: SDDB record 52, loop L010 (01:16–01:25), $N_H = 413$, about 70 minutes before the VF onset.

5. Hypothesis

A pathology does not come suddenly. If the structure of the heart is lost before an event, the loss should be visible in the entropy cube, loop after loop, before the event.

This is a hypothesis. Record 52 is a single recording of a single patient, and it is at 250 Hz.

6. Proposed work

1. **SDDB records 30, 45 and 51**, which have long recordings before the VF onset, learned in successive loops as record 52.
2. **Standard recordings.** 1000 Hz, lead II, runs long enough for saturation, to build a library of entropy cubes of healthy hearts and of hearts with a known condition.
3. **Monitoring.** A small lead II capture device with three electrodes, and a mobile phone hosting the database and the dashboards. The phone sends the events of each loop ($\mathrm{Re}\,\dot{c}$, Δt), not the raw ECG, to a remote SKA engine API, and receives the loop's cube. One loop per hour or per day, so that each heart is compared with itself over time.
4. **Other physiological signals.** The ECG is one parameter of the heart. Blood pressure, recorded continuously with the ECG in the Autonomic Aging database, can be read the same way.

7. What SKA does not do

SKA is not a classifier and gives no diagnosis. It does not produce a risk score and is not trained on labels. It registers the structure of the stream it reads; what that structure means for a patient is for cardiologists to establish.

8. Collaboration

We seek cardiologists and research groups with access to long ECG recordings, ideally at 1000 Hz, or to patients followed over time, to read the cubes with us and to establish what the loss of structure corresponds to.

9. Repositories

All results, figures and cubes are public. The cubes open in a browser or in a VR headset. The SKA real-time engine is proprietary and is not included.

- SKA-ECG-Heart-Rate-Variability: https://github.com/quantiota/SKA-ECG-Heart-Rate-Variability
- SKA-ECG-Pathology: https://github.com/quantiota/SKA-ECG-Pathology
- SKA-ECG-SDDB: https://github.com/quantiota/SKA-ECG-SDDB
- SKA-ECG-Fantasia: https://github.com/quantiota/SKA-ECG-Fantasia
- SKA-ECG-Athletes: https://github.com/quantiota/SKA-ECG-Athletes
- SKA-ECG-Detection-Validation: https://github.com/quantiota/SKA-ECG-Detection-Validation
- SKA-ECG-Arrhythmia: https://github.com/quantiota/SKA-ECG-Arrhythmia
- SKA-ECG-Arrhythmia-Second-Order (record 119): https://github.com/quantiota/SKA-ECG-Arrhythmia-Second-Order
- SKA-ECG-Second-Order, SKA-ECG-Third-Order: https://github.com/quantiota

References

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